UNIVERSITY OF WOLLONGONG AUSTRALIA

Tutorial 01

Physical models and independent states

Preparation: Lecture 01: system boundaries, first-principles balances, constitutive laws and model order.

Submission credit: 1% of the course grade for submitted work; correctness is not marked.

Show essential substitutions and units.

Question 1 Concepts and a first-order model

Give a brief reason for each conceptual answer; show the mass balance and substitutions for the pressure vessel.

a) Classify the ideal resistor, one well-mixed heated room, and one mass attached to a fixed support by a spring and damper as static or dynamic. The room has constant heat capacity and a fixed ambient temperature. For each dynamic model, name the minimum state variables and give its order.

b) Model classification. Classify each equation as linear time invariant, linear time varying, or nonlinear. The unknown is x(t); primes denote time derivatives. Give the feature that decides your classification.

$(i)\quad x'' + 3x' + 2x = \sin(t)$

$(ii)\quad x'' + (2 + \cos(t))x = 0$

$(iii)\quad x'' + x^3 = 0$

c) Figure 1 shows a mass between fixed walls, connected by springs k_1 = 120 N/m and k_2 = 180 N/m. For a rightward displacement x from equilibrium, find the equivalent stiffness. Justify the spring combination from the restoring forces.

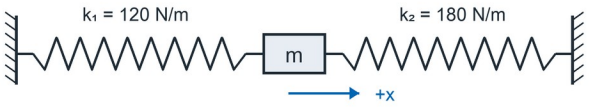


Figure 1. Springs attached to the same moving mass.

d) Air enters a rigid isothermal vessel through one restrictive valve; there is no outlet (Lecture 1, slide 25). Supply and vessel pressure deviations are p_i and p_o; inlet mass flow is q. The linear laws are $Rq = p_i - p_o$ and $dm = C\,dp_o$, with $R = 1.0 \times 10^9$ Pa·s/kg and $C = 1.0 \times 10^{-8}$ kg/Pa.

Use a mass balance to derive the ODE. Identify input, state, output, order and time constant. Include units.

Question 2 A packaging cart coupled to a pinion

Use an effective translating mass of 5 kg for the cart and ideal rack–pinion mechanism, including the pinion inertia. Two parallel 600 N/m springs are in series with a massless 300 N/m mount. A 24 N s/m damper connects the cart to ground. Force F and displacement x are positive rightward from equilibrium.

$m_{eff} = 5\,kg$

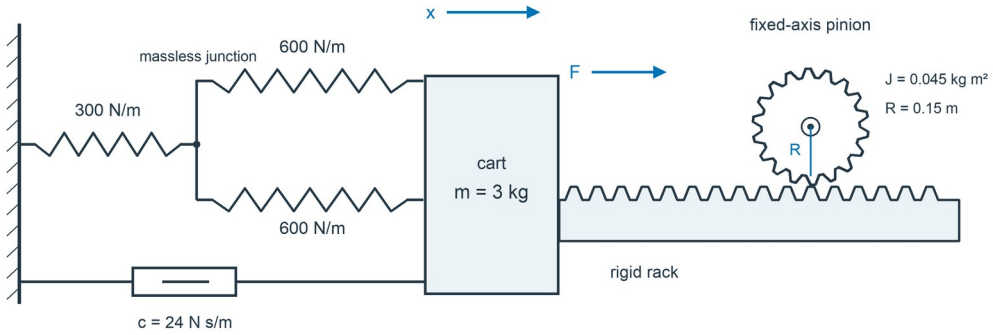


Figure 2. Cart and spring assembly with supplied effective mass.

a) Reduce the spring assembly in **Figure 2**. Use the supplied effective mass to write the numerical equation of motion.

b) For a 12 N force step from rest, find the initial acceleration and final displacement.

Question 3 Two connected heated enclosures

Two well-mixed enclosures exchange heat with each other and with a fixed ambient. The temperature rises above ambient are θ_1 and θ_2 (°C); P_1 and P_2 are independent heater powers (W). Use the numerical balances below, with time in seconds and a dot denoting a time derivative. Heat capacities and conductances are constant.

$$1200\,\dot{\theta}_1 = P_1 - 3\,\theta_1 - 6\,(\theta_1 - \theta_2)$$

$$1800\,\dot{\theta}_2 = P_2 + 6\,(\theta_1 - \theta_2) - 4\,\theta_2$$

a) Identify the two heater inputs, the two temperature states and outputs, and the model order from the supplied balances.

b) Evaluate both temperature rates when θ_1 = 10°C, θ_2 = 0°C, P_1 = 60 W and P_2 = 0 W. Can an enclosure cool while its heater is on?

Hint

Method source: Ogata, System Dynamics, 4th ed., Problem A-3-12, pp. 90–91 (mechanical constraints); Modern Control Engineering, 5th ed., Section 4-5 (thermal balances). The scenarios and data in this tutorial are new adaptations. Page numbers refer to printed pages.

Prof. Zengxi (Stephen) Pan | University of Wollongong | TYUT 2026

October 2026